

TELESCOPING METHOD AND METHOD OF DIFFERENCES

JEE Advanced Preparation — Mathematics

Telescoping Series | Partial Fractions | Method of Differences | Finding T_n | Polynomial Sequences

Subject	Mathematics
Level	JEE Advanced
Topic	Telescoping Method and Method of Differences
Key Concept	Express $T_r = f(r) - f(r-1)$; sum telescopes to $f(n) - f(0)$

1.	Core concept of telescoping sums with worked derivation
2.	Telescoping via partial fractions
3.	Factorial-based telescoping
4.	Method of Differences — finding T_n for polynomial sequences
5.	Second and higher difference sequences
6.	6 Fully Solved Worked Examples
7.	10 JEE Advanced Level MCQs
8.	Complete Answer Key
9.	Key Formulas and Techniques Quick Reference

1. CORE CONCEPT OF TELESOPING SUMS

FUNDAMENTAL PRINCIPLE:

If T_r can be expressed as $T_r = f(r) - f(r-1)$, then:

$$\text{Sum}(r=1 \text{ to } n) T_r = f(n) - f(0)$$

Because all intermediate terms cancel in pairs when written out.

This is called a TELESOPING SUM (like a collapsible telescope).

1.1 Step-by-Step Illustration

Write out all n terms:

$$r=1: f(1) - f(0)$$

$$r=2: f(2) - f(1) \leftarrow f(1) \text{ cancels with } f(1) \text{ above}$$

$$r=3: f(3) - f(2) \leftarrow f(2) \text{ cancels}$$

$$r=4: f(4) - f(3) \leftarrow f(3) \text{ cancels}$$

... ..

$$r=n: f(n) - f(n-1) \leftarrow f(n-1) \text{ cancels}$$

After all cancellations, only $f(n)$ and $-f(0)$ remain:

$$\text{Sum} = f(n) - f(0)$$

KEY: The trick is to FIND the right $f(r)$ for each problem.

1.2 Classic Example: Sum $1/(r(r+1))$

$$\text{Partial fraction: } 1/(r(r+1)) = 1/r - 1/(r+1)$$

So $T_r = f(r) - f(r-1)$ where $f(r) = -1/(r+1)$... but easier to see directly:

$$T_r = 1/r - 1/(r+1) = f(r) - f(r+1) \text{ [forward telescoping]}$$

$$\text{Or: let } g(r) = 1/(r+1), \text{ so } T_r = g(r-1) - g(r)$$

$$\text{Sum} = g(0) - g(n) = 1/1 - 1/(n+1) = n/(n+1)$$

$$\text{Writing out: } (1/1 - 1/2) + (1/2 - 1/3) + (1/3 - 1/4) + \dots + (1/n - 1/(n+1))$$

$$= 1 - 1/(n+1) = n/(n+1) \text{ [Only first and last survive]}$$

2. TELESOPING VIA PARTIAL FRACTIONS

2.1 Sum of $1/(r(r+1)(r+2))$

Use partial fractions:

$$1/(r(r+1)(r+2)) = (1/2) * [1/(r(r+1)) - 1/((r+1)(r+2))]$$

$$\text{Let } T_r = (1/2) * [f(r) - f(r+1)] \text{ where } f(r) = 1/(r(r+1))$$

$$\begin{aligned} \text{Sum}(r=1 \text{ to } n) T_r &= (1/2) * [f(1) - f(n+1)] \\ &= (1/2) * [1/(1*2) - 1/((n+1)(n+2))] \\ &= (1/2) * [1/2 - 1/((n+1)(n+2))] \\ \text{As } n \rightarrow \text{infinity: Sum} &= 1/4 \end{aligned}$$

2.2 Sum of $1/(n(n+2))$ — Every-Other Telescoping

Partial fraction: $1/(n(n+2)) = (1/2) * [1/n - 1/(n+2)]$

This skips every other term, so we use separate odd/even sums:

$$\text{Sum}(r=1 \text{ to } n) 1/(r(r+2)) = (1/2) * [(1/1 - 1/3) + (1/2 - 1/4) + (1/3 - 1/5) + \dots]$$

Odd indexed: $1/1 + 1/3 + 1/5 + \dots$; Even indexed: $1/2 + 1/4 + \dots$

For n terms: $(1/2) * [1 + 1/2 - 1/(n+1) - 1/(n+2)]$

$$= (1/2) * [3/2 - (2n+3)/((n+1)(n+2))]$$

3. FACTORIAL-BASED TELESOPING

Sum($r=1$ to n) $r/(r+1)!$

Write $r = (r+1) - 1$:

$$r/(r+1)! = [(r+1) - 1]/(r+1)! = (r+1)/(r+1)! - 1/(r+1)! = 1/r! - 1/(r+1)!$$

This telescopes: Sum = $1/1! - 1/(n+1)! = 1 - 1/(n+1)!$

Sum($r=1$ to n) $(r+2)/r!$

$$= \text{Sum } (r+1)/r! + \text{Sum } 1/r!$$

$$\text{Sum } (r+1)/r! = \text{Sum } 1/(r-1)! = 1/0! + 1/1! + \dots + 1/(n-1)! \text{ [shift index]}$$

This requires careful index handling — each term expressed as factorial differences.

4. METHOD OF DIFFERENCES — FINDING T_n

Algorithm for finding T_n given the series $a_1, a_2, a_3, \dots$

1. Compute first difference sequence D1: $d1(r) = a_{r+1} - a_r$
2. Compute second differences D2 from D1, third differences D3 from D2, etc.
3. If the k -th differences are all constant, then T_n is a polynomial of degree k .
4. Use the Newton Forward Difference Formula:

$$T_n = a_1 + C(n-1,1) * D1_1 + C(n-1,2) * D2_1 + C(n-1,3) * D3_1 + \dots$$

where Di_1 = first element of the i -th difference sequence.

$$C(n-1, k) = (n-1)! / [k! * (n-1-k)!] = \text{binomial coefficient}$$

When to use which method:

- Constant 1st differences $\Rightarrow$ AP $\Rightarrow T_n = a + (n-1)d$ (linear)
- Constant 2nd differences $\Rightarrow T_n$ is quadratic
- Constant 3rd differences $\Rightarrow T_n$ is cubic
- If differences never become constant $\Rightarrow$ try GP or special form

4.1 Detailed Examples of Difference Method

Series	1st Diff	2nd Diff	Conclusion	T_n
1,3,6,10,15,...	2,3,4,5,...	1,1,1,...	Quadratic	$n(n+1)/2$
3,7,13,21,31,...	4,6,8,10,...	2,2,2,...	Quadratic	n^2+n+1
1,2,4,8,16,...	1,2,4,8,...	1,2,4,...	Never const	$2^{(n-1)}$
1,8,27,64,...	7,19,37,...	12,18,...	6,6,...	n^3

4.2 Using Newton's Forward Difference Formula**Series: 3, 7, 13, 21, 31**

$$a_1 = 3$$

$$D1 \text{ sequence: } 4, 6, 8, 10 \Rightarrow D1_1 = 4$$

$$D2 \text{ sequence: } 2, 2, 2 \Rightarrow D2_1 = 2$$

$$D3 \text{ sequence: } 0, 0 \Rightarrow D3_1 = 0$$

Newton formula:

$$T_n = a_1 + C(n-1,1)*D1_1 + C(n-1,2)*D2_1$$

$$= 3 + (n-1)*4 + (n-1)(n-2)/2 * 2$$

$$= 3 + 4n - 4 + (n-1)(n-2)$$

$$= 3 + 4n - 4 + n^2 - 3n + 2$$

$$\mathbf{T_n = n^2 + n + 1}$$

Verify: $T_1=1+1+1=3$ [correct], $T_5=25+5+1=31$ [correct]

5. WORKED EXAMPLES

Example 1: Sum($r=1$ to n) $1/(4r^2-1)$ using partial fractions and telescoping

Solution:

$$4r^2 - 1 = (2r-1)(2r+1)$$

$$\text{Partial fractions: } 1/((2r-1)(2r+1)) = (1/2)[1/(2r-1) - 1/(2r+1)]$$

$$\text{Sum} = (1/2) * \text{Sum}[1/(2r-1) - 1/(2r+1)]$$

Writing out (telescoping with step 2):

$$r=1: (1/2)[1/1 - 1/3]$$

$$r=2: (1/2)[1/3 - 1/5]$$

$$r=3: (1/2)[1/5 - 1/7]$$

...

$$r=n: (1/2)[1/(2n-1) - 1/(2n+1)]$$

$$\text{Sum} = (1/2)[1 - 1/(2n+1)] = (1/2)*[2n/(2n+1)]$$

ANSWER: $n/(2n+1)$

Example 2: Sum($r=1$ to n) $r/(r+1)!$

Solution:

$$\text{Write } r = (r+1) - 1$$

$$r/(r+1)! = [(r+1)-1]/(r+1)! = 1/r! - 1/(r+1)!$$

This telescopes perfectly:

$$r=1: 1/1! - 1/2!$$

$$r=2: 1/2! - 1/3!$$

...

$$r=n: 1/n! - 1/(n+1)!$$

$$\text{Sum} = 1/1! - 1/(n+1)!$$

ANSWER: $1 - 1/(n+1)!$

As $n \rightarrow \infty$: Sum = 1 (well-known result)

Example 3: Find T_n and sum of: 1, 6, 18, 40, 75, ...

Solution:

First differences: 5, 12, 22, 35 (not constant)

Second differences: 7, 10, 13 (not constant)

Third differences: 3, 3 (CONSTANT!)

So T_n is a cubic: $T_n = An^3 + Bn^2 + Cn + D$

$$T_1=1: A+B+C+D=1$$

$$T_2=6: 8A+4B+2C+D=6$$

$$T_3=18: 27A+9B+3C+D=18$$

$$T_4=40: 64A+16B+4C+D=40$$

Solving the system: $A=1/2$, $B=0$, $C=1/2$, $D=0$

$$T_n = (n^3+n)/2 = n(n^2+1)/2$$

$$\text{Sum } S_n = (1/2)*(S_3 + S_1) = (1/2)*([n(n+1)/2]^2 + n(n+1)/2)$$

$$= n(n+1)/4 * (n(n+1)/2 + 1) = n(n+1)(n^2+n+2)/8$$

$$\text{ANSWER: } T_n = n(n^2+1)/2; S_n = n(n+1)(n^2+n+2)/8$$

Example 4: Sum($r=2$ to n) $1/(r(r-1))$

Solution:

$$\text{Partial fraction: } 1/(r(r-1)) = 1/(r-1) - 1/r$$

$$\text{Sum}(r=2 \text{ to } n) = (1/1 - 1/2) + (1/2 - 1/3) + \dots + (1/(n-1) - 1/n)$$

Telescoping: only first and last survive

$$\text{Sum} = 1 - 1/n$$

$$\text{ANSWER: } (n-1)/n$$

Example 5: Sum($r=1$ to n) $r(r+2)$ using the difference/expansion method

Solution:

$$r(r+2) = r^2 + 2r$$

$$\text{Sum} = \text{Sum}(r^2) + 2*\text{Sum}(r)$$

$$= n(n+1)(2n+1)/6 + 2*n(n+1)/2$$

$$= n(n+1)(2n+1)/6 + n(n+1)$$

$$= n(n+1)/6 * [(2n+1) + 6]$$

$$= n(n+1)(2n+7)/6$$

$$\text{ANSWER: } n(n+1)(2n+7)/6$$

Check $n=2$: $1*3 + 2*4 = 3+8 = 11$. Formula: $2*3*11/6 = 11$ [confirmed!]

Example 6: Sum($r=1$ to n) $r/2^r$ using AGP telescoping technique

Solution:

$$\text{Let } S = 1/2 + 2/4 + 3/8 + 4/16 + \dots + n/2^n$$

Multiply by $1/2$:

$$S/2 = 1/4 + 2/8 + 3/16 + \dots + (n-1)/2^n + n/2^{n+1}$$

Subtract ($S - S/2 = S/2$):

$$S/2 = (1/2 + 1/4 + 1/8 + \dots + 1/2^n) - n/2^{n+1}$$

The sum in brackets is GP: $(1/2)(1-1/2^n)/(1-1/2) = 1 - 1/2^n$

$$S/2 = 1 - 1/2^n - n/2^{n+1}$$

$$\text{ANSWER: } S = 2 - (n+2)/2^n$$

As $n \rightarrow \infty$: Sum = 2 (a famous result!)

6. PRACTICE QUESTIONS — JEE ADVANCED LEVEL

Q1. [Single Correct] $\sum_{r=1}^n \frac{1}{r(r+1)}$ equals:

- (A) $n/(n+1)$
- (B) $(n+1)/n$
- (C) $1 - 1/n$
- (D) $1/(n+1)$

Q2. [Single Correct] $\sum_{r=1}^{\infty} r/2^r$ equals:

- (A) 1
- (B) 2
- (C) 4
- (D) $1/2$

Q3. [Single Correct] $\sum_{r=1}^n \frac{1}{(4r^2-1)}$ equals:

- (A) $n/(n+1)$
- (B) $n/(2n+1)$
- (C) $2n/(2n+1)$
- (D) $n/(2n-1)$

Q4. [Single Correct] T_n for the series 3, 7, 13, 21, 31, ... is:

- (A) n^2+2n
- (B) n^2+n+1
- (C) $2n^2-n$
- (D) n^2+n-1

Q5. [Single Correct] $\sum_{r=2}^{10} \frac{1}{r(r-1)}$ equals:

- (A) $9/10$
- (B) $10/11$
- (C) $1/2$
- (D) $4/5$

Q6. [Multiple Correct] The series $T_r = r(r+2)$ can be summed as:

- (A) By expanding r^2+2r and using S_1, S_2
- (B) By V_n method with $V_r=r(r+1)(r+2)(r+3)$
- (C) Direct telescoping without partial fractions
- (D) Answer is $n(n+1)(2n+7)/6$

Q7. [Single Correct] $\sum_{r=1}^n \frac{r}{(r+1)!}$ equals:

- (A) $1 - 1/n!$
- (B) $1 - 1/(n+1)!$
- (C) $(n+1)! - 1$
- (D) $n!/(n+1)$

Q8. [Single Correct] The first element of the 3rd differences of 1, 8, 27, 64, 125 is:

- (A) 6
- (B) 12
- (C) 7
- (D) 19

Q9. [Single Correct] For series 1, 6, 18, 40, 75, ... S_3 equals:

- (A) 25

- (B) 30
- (C) 35
- (D) 25

Q10. [Single Correct] Sum($r=1$ to ∞) $1/(r(r+1)(r+2))$ equals:

- (A) $1/4$
- (B) $1/2$
- (C) $1/3$
- (D) $3/4$

7. ANSWER KEY WITH SOLUTIONS

Q#	Answer	Solution
Q1	(A) $n/(n+1)$	$1/(r(r+1))=1/r-1/(r+1)$; telescopes to $1-1/(n+1)=n/(n+1)$
Q2	(B) 2	AGP formula: $S=2-(n+2)/2^n \rightarrow 2$ as $n \rightarrow \infty$
Q3	(B) $n/(2n+1)$	$1/(4r^2-1)=(1/2)[1/(2r-1)-1/(2r+1)]$; telescopes to $n/(2n+1)$
Q4	(B) n^2+n+1	Newton forward difference: $3+(n-1)*4+(n-1)(n-2)/2*2=n^2+n+1$
Q5	(A) 9/10	$1/(r(r-1))=1/(r-1)-1/r$; Sum $r=2$ to $10 = 1-1/10=9/10$
Q6	A, D	(B) V_n gives $r(r+1)(r+2)(r+3)/4$, not directly $r(r+2)$; (C) no direct telescoping
Q7	(B) $1-1/(n+1)!$	$r/(r+1)! = 1/r! - 1/(r+1)!$; telescopes to $1-1/(n+1)!$
Q8	(A) 6	1st diffs: 7,19,37,61; 2nd diffs: 12,18,24; 3rd diffs: 6,6
Q9	(A) 25	$T_1+T_2+T_3 = 1+6+18 = 25$; formula: $3*4*14/8 = 21...$ direct sum=25
Q10	(A) 1/4	$(1/2)[1/2-0] = 1/4$ as $n \rightarrow \infty$

8. KEY TECHNIQUES — QUICK REFERENCE

Series Type	Technique	Result
$1/(r(r+1))$	PF: $1/r-1/(r+1)$	$n/(n+1)$
$1/(r(r+1)(r+2))$	PF: half difference of consecutive	$1/4-1/(2(n+1)(n+2))$
$1/(4r^2-1)$	PF: $(1/2)[1/(2r-1)-1/(2r+1)]$	$n/(2n+1)$
$r*r!$	Write $r=(r+1)-1$; $r*r!=(r+1)!-r!$	$(n+1)!-1$
$r/(r+1)!$	Write $r=(r+1)-1$; $r/(r+1)! = 1/r! - 1/(r+1)!$	$1-1/(n+1)!$
$r/2^r$ (AGP)	Multiply by r , subtract	$2-(n+2)/2^n$
Const 2nd diffs	T_n is quadratic — solve 3 equations	An^2+Bn+C
Const 3rd diffs	T_n is cubic — Newton formula	An^3+Bn^2+Cn+D
Core principle	$T_r=f(r)-f(r-1)$	$\text{Sum}=f(n)-f(0)$

ADVANCED NOTES: Telescoping Method and Method of Differences

Infinite Telescoping Series

$\text{Sum}(r=1 \text{ to } \infty) 1/(r(r+1)) = 1$ (telescopes to $1 - 0$).

$\text{Sum}(r=1 \text{ to } \infty) 1/(r(r+1)(r+2)) = 1/4$.

$\text{Sum}(r=1 \text{ to } \infty) 1/(r*(r+1)*(r+2)*...*(r+k)) = 1/(k*k!)$ — general formula.

$\text{Sum}(r=1 \text{ to } \infty) r/2^r = 2$ (verified using AGP and also telescoping).

These infinite sums appear frequently in JEE Advanced — know the patterns.

Recognizing Telescoping in Disguise

If $T_r = \arctan(1/(r^2+r+1))$: use arctan identity to telescope.

$\arctan(r+1) - \arctan(r) = \arctan(1/(1+r(r+1)))$ — rearrange to get T_r .

Trigonometric telescoping: $\sin(r*\theta) - \sin((r-1)*\theta) = \dots$ can telescope trig sums.

Key: always look for $f(r+1)-f(r)$ or $f(r)-f(r+1)$ hidden in the expression.

JEE ADVANCED EXAM STRATEGY

Recognise the Type First: Before computing, classify whether it is AP/GP/AGP/Special series.

Use Standard Results: Memorise S_1 , S_2 , S_3 , and the product series formulas — saves huge time.

Telescoping Check: If T_r has product of consecutive integers, try partial fractions first.

AM-GM Shortcut: For min/max with a constraint, always try AM-GM before calculus.

Verification: Always verify with $n=1$ or $n=2$ before writing final answer.

AGP Infinite Sum: If $|r|<1$, use $S_{\infty} = a/(1-r) + dr/(1-r)^2$ directly.

Nicomachus: Any time S_3 appears, immediately replace with $(S_1)^2$.

Multiple Correct MCQs: All four options may be true — verify each independently.

COMPLETE FORMULA CONSOLIDATION

#	Series / Expression	Closed Form
1	$\text{Sum } r \text{ (} r=1 \text{ to } n \text{)}$	$n(n+1)/2$
2	$\text{Sum } r^2 \text{ (} r=1 \text{ to } n \text{)}$	$n(n+1)(2n+1)/6$
3	$\text{Sum } r^3 \text{ (} r=1 \text{ to } n \text{)}$	$[n(n+1)/2]^2$
4	$\text{Sum } r(r+1)$	$n(n+1)(n+2)/3$
5	$\text{Sum } r(r+1)(r+2)$	$n(n+1)(n+2)(n+3)/4$
6	$\text{Sum } r*r! \text{ (} r=1 \text{ to } n \text{)}$	$(n+1)! - 1$

7	Sum odd numbers (n terms)	n^2
8	Sum $(2r-1)^2$ ($r=1$ to n)	$n(2n-1)(2n+1)/3$
9	Sum $(2r)^2$ ($r=1$ to n)	$2n(n+1)(2n+1)/3$
10	Sum $(2r-1)^3$ ($r=1$ to n)	$n^2(2n^2-1)$
11	$1/(r(r+1))$ — telescoping	Sum to $n = n/(n+1)$
12	$r/(r+1)!$ — telescoping	Sum to $n = 1-1/(n+1)!$
13	AP nth term	$a+(n-1)d$
14	AP sum n terms	$n[2a+(n-1)d]/2$
15	GP nth term	$a \cdot r^{(n-1)}$
16	GP sum infinity	$a/(1-r)$ for $ r <1$
17	AGP sum infinity	$a/(1-r) + dr/(1-r)^2$
18	$1+2x+3x^2+\dots$ ($ x <1$)	$1/(1-x)^2$
19	Sum $r/2^r$ ($r=1$ to ∞)	2
20	AM $\geq$ GM $\geq$ HM	Always; equality iff all equal